PHYS4055 Week 5 Tutorial Solutions

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(Dated: August 24, 2026)

Worked Solution: A Pure-Loss Channel and Conditional Evolution

1. **Joint system–environment state**

The initial joint state is

$$|\Psi_{\text{in}}\rangle_{SE} = (\alpha|0\rangle_S + \beta|1\rangle_S)\,|0\rangle_E. \tag{1}$$

Applying the beam-splitter transformations to each term gives

$$|\Psi_{\text{out}}\rangle_{SE} = \alpha|0\rangle_S|0\rangle_E + \sqrt{\eta}\,\beta|1\rangle_S|0\rangle_E + \sqrt{1-\eta}\,\beta|0\rangle_S|1\rangle_E \tag{2}$$

$$= (\alpha|0\rangle_S + \sqrt{\eta}\,\beta|1\rangle_S)\,|0\rangle_E + \sqrt{1-\eta}\,\beta|0\rangle_S|1\rangle_E. \tag{3}$$

The first term corresponds to no photon being found in the environment, while the second corresponds to the photon being lost into the environment.

2. **Kraus operators**

The Kraus operators are defined by

$$\hat{K}_j = {}_E\langle j|\hat{U}_\eta|0\rangle_E. \tag{4}$$

We first determine how $\hat{K}_0$ acts on the system basis states:

$$\hat{K}_0|0\rangle_S = {}_E\langle 0|\hat{U}_\eta|0\rangle_S|0\rangle_E = |0\rangle_S, \tag{5}$$

$$\hat{K}_0|1\rangle_S = {}_E\langle 0|\hat{U}_\eta|1\rangle_S|0\rangle_E = \sqrt{\eta}\,|1\rangle_S. \tag{6}$$

Therefore,

$$\boxed{\hat{K}_0 = |0\rangle\langle 0| + \sqrt{\eta}\,|1\rangle\langle 1|}. \tag{7}$$

Similarly,

$$\hat{K}_1|0\rangle_S = {}_E\langle 1|\hat{U}_\eta|0\rangle_S|0\rangle_E = 0, \tag{8}$$

$$\hat{K}_1|1\rangle_S = {}_E\langle 1|\hat{U}_\eta|1\rangle_S|0\rangle_E = \sqrt{1-\eta}\,|0\rangle_S. \tag{9}$$

Thus,

$$\boxed{\hat{K}_1 = \sqrt{1-\eta}\,|0\rangle\langle 1|}. \tag{10}$$

In the ordered basis $\{|0\rangle, |1\rangle\}$, these operators are

$$\hat{K}_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{\eta} \end{pmatrix}, \qquad \hat{K}_1 = \begin{pmatrix} 0 & \sqrt{1-\eta} \\ 0 & 0 \end{pmatrix}. \tag{11}$$

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We now check the completeness relation:

$$\hat{K}_0^\dagger \hat{K}_0 = |0\rangle\langle 0| + \eta|1\rangle\langle 1|, \tag{12}$$
$$\hat{K}_1^\dagger \hat{K}_1 = (1-\eta)|1\rangle\langle 1|. \tag{13}$$

Therefore,

$$\hat{K}_0^\dagger \hat{K}_0 + \hat{K}_1^\dagger \hat{K}_1 = |0\rangle\langle 0| + [\eta + (1-\eta)]\,|1\rangle\langle 1| \tag{14}$$
$$= |0\rangle\langle 0| + |1\rangle\langle 1| \tag{15}$$
$$= \hat{I}_S. \tag{16}$$

Consequently,

$$\mathrm{Tr}(\hat{\rho}'_S) = \mathrm{Tr}\left(\sum_{j=0}^{1} \hat{K}_j \hat{\rho}_S \hat{K}_j^\dagger\right) \tag{17}$$
$$= \mathrm{Tr}\left[\hat{\rho}_S \sum_{j=0}^{1} \hat{K}_j^\dagger \hat{K}_j\right] \tag{18}$$
$$= \mathrm{Tr}(\hat{\rho}_S) = 1. \tag{19}$$

The completeness relation therefore ensures that the complete channel preserves the trace.

3. **Conditional evolution**

For the environmental outcome $j = 0$, the unnormalised conditional system state vector is

$$\hat{K}_0|\psi\rangle_S = \alpha|0\rangle_S + \sqrt{\eta}\,\beta|1\rangle_S. \tag{20}$$

Its probability is

$$p_0 = \langle\psi|\hat{K}_0^\dagger \hat{K}_0|\psi\rangle \tag{21}$$
$$= |\alpha|^2 + \eta|\beta|^2. \tag{22}$$

The corresponding normalised conditional state is

$$\boxed{|\psi_0\rangle_S = \frac{\alpha|0\rangle_S + \sqrt{\eta}\,\beta|1\rangle_S}{\sqrt{|\alpha|^2 + \eta|\beta|^2}}}. \tag{23}$$

This is the no-loss outcome: no photon is found in the environment.

For the outcome $j = 1$,

$$\hat{K}_1|\psi\rangle_S = \sqrt{1-\eta}\,\beta|0\rangle_S. \tag{24}$$

The probability of this outcome is

$$p_1 = \langle\psi|\hat{K}_1^\dagger \hat{K}_1|\psi\rangle \tag{25}$$
$$= (1-\eta)|\beta|^2. \tag{26}$$

Provided that $p_1 \neq 0$, the normalised conditional state is, up to an irrelevant global phase,

$$\boxed{|\psi_1\rangle_S = |0\rangle_S}. \tag{27}$$

This outcome corresponds to photon loss: one photon is found in the environment and the system is left in the vacuum state.

The two probabilities sum to

$$p_0 + p_1 = |\alpha|^2 + \eta|\beta|^2 + (1-\eta)|\beta|^2 \tag{28}$$
$$= |\alpha|^2 + |\beta|^2 = 1. \tag{29}$$

An individual conditional operation is not trace-preserving because

$$\mathrm{Tr}\left(\hat{K}_j \hat{\rho}_S \hat{K}_j^\dagger\right) = p_j, \tag{30}$$

which is generally less than one. The trace is the probability that outcome j occurs. The remaining probability corresponds to the other possible environmental outcomes.

4. **Unconditional reduced system state**

The input density operator is

$$\hat{\rho}_S = |\psi\rangle\langle\psi| \tag{31}$$
$$= |\alpha|^2|0\rangle\langle 0| + \alpha\beta^*|0\rangle\langle 1| + \alpha^*\beta|1\rangle\langle 0| + |\beta|^2|1\rangle\langle 1|. \tag{32}$$

In matrix form,

$$\hat{\rho}_S = \begin{pmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{pmatrix}. \tag{33}$$

The no-loss contribution is

$$\hat{K}_0 \hat{\rho}_S \hat{K}_0^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{\eta} \end{pmatrix} \begin{pmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{\eta} \end{pmatrix} \tag{34}$$
$$= \begin{pmatrix} |\alpha|^2 & \sqrt{\eta}\,\alpha\beta^* \\ \sqrt{\eta}\,\alpha^*\beta & \eta|\beta|^2 \end{pmatrix}. \tag{35}$$

The loss contribution is

$$\hat{K}_1 \hat{\rho}_S \hat{K}_1^\dagger = \begin{pmatrix} 0 & \sqrt{1-\eta} \\ 0 & 0 \end{pmatrix} \begin{pmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ \sqrt{1-\eta} & 0 \end{pmatrix} \tag{36}$$
$$= \begin{pmatrix} (1-\eta)|\beta|^2 & 0 \\ 0 & 0 \end{pmatrix}. \tag{37}$$

Adding the two possible outcomes gives

$$\boxed{\hat{\rho}'_S = \begin{pmatrix} |\alpha|^2 + (1-\eta)|\beta|^2 & \sqrt{\eta}\,\alpha\beta^* \\ \sqrt{\eta}\,\alpha^*\beta & \eta|\beta|^2 \end{pmatrix}}. \tag{38}$$

Equivalently,

$$\hat{\rho}'_S = \left[|\alpha|^2 + (1-\eta)|\beta|^2\right]|0\rangle\langle 0| + \eta|\beta|^2|1\rangle\langle 1| + \sqrt{\eta}\,\alpha\beta^*|0\rangle\langle 1| + \sqrt{\eta}\,\alpha^*\beta|1\rangle\langle 0|. \tag{39}$$

The one-photon population is multiplied by η because η is the probability that the photon is transmitted. The corresponding probability amplitude is multiplied by $\sqrt{\eta}$, so the coherences between the vacuum and one-photon components are multiplied by $\sqrt{\eta}$.

Conditioning on a particular environmental outcome selects a single pure branch of the joint state. If the environmental outcome is ignored, the no-loss and loss branches must be added incoherently. Because the environment contains information about which process occurred, the resulting reduced system state can be mixed.

5. **Purity of the output state**

Using the normalisation condition,

$$|\alpha|^2 = 1 - |\beta|^2, \tag{40}$$

the output matrix elements may be written as

$$\rho'_{00} = 1 - \eta|\beta|^2, \tag{41}$$
$$\rho'_{11} = \eta|\beta|^2, \tag{42}$$
$$|\rho'_{01}|^2 = \eta|\alpha|^2|\beta|^2. \tag{43}$$

For a two-dimensional density matrix,

$$\mathrm{Tr}\left[(\hat{\rho}'_S)^2\right] = (\rho'_{00})^2 + (\rho'_{11})^2 + 2\,|\rho'_{01}|^2\,. \tag{44}$$

Substitution gives

$$\mathrm{Tr}\left[(\hat{\rho}'_S)^2\right] = \left(1 - \eta|\beta|^2\right)^2 + \eta^2|\beta|^4 + 2\eta|\alpha|^2|\beta|^2 \tag{45}$$
$$= \left(1 - \eta|\beta|^2\right)^2 + \eta^2|\beta|^4 + 2\eta\left(1 - |\beta|^2\right)|\beta|^2 \tag{46}$$
$$= 1 - 2\eta|\beta|^2 + \eta^2|\beta|^4 + \eta^2|\beta|^4 + 2\eta|\beta|^2 - 2\eta|\beta|^4 \tag{47}$$
$$= \boxed{1 - 2\eta(1-\eta)|\beta|^4}. \tag{48}$$

For $0 < \eta < 1$, the factor $\eta(1-\eta)$ is nonzero. Therefore, the output has unit purity only when

$$|\beta| = 0. \tag{49}$$

Thus, within this range of η, only the vacuum input

$$|\psi\rangle_S = |0\rangle_S \tag{50}$$

remains pure.

At $\eta = 1$, the channel is the identity and every pure input remains pure. At $\eta = 0$, every input is mapped to the vacuum state, which is also pure.

For $0 < \eta < 1$ and $|\beta| \neq 0$, the system becomes correlated with the environment. Since the total system–environment state remains pure under the unitary evolution, the mixedness of the reduced system state is a consequence of system–environment entanglement.

6. **Photon-number statistics for a one-photon input**

For the input state

$$|\psi\rangle_S = |1\rangle_S, \tag{51}$$

we have $\alpha = 0$ and $\beta = 1$. The output state becomes

$$\hat{\rho}'_S = (1-\eta)|0\rangle\langle 0| + \eta|1\rangle\langle 1|. \tag{52}$$

Therefore,

$$P(n=0) = 1 - \eta, \qquad P(n=1) = \eta. \tag{53}$$

Within the single-rail subspace,

$$\hat{n} = |1\rangle\langle 1|, \tag{54}$$

and hence

$$\hat{n}^2 = \hat{n}. \tag{55}$$

The mean photon number is

$$\begin{aligned} \langle \hat{n} \rangle &= \mathrm{Tr}\left(\hat{n}\hat{\rho}'_S\right) && (56)\\ &= 0 \times (1-\eta) + 1 \times \eta && (57)\\ &= \eta. && (58) \end{aligned}$$

Similarly,

$$\begin{aligned} \langle \hat{n}^2 \rangle &= \mathrm{Tr}\left(\hat{n}^2\hat{\rho}'_S\right) && (59)\\ &= \eta. && (60) \end{aligned}$$

The variance is therefore

$$\begin{aligned} \mathrm{Var}(\hat{n}) &= \langle \hat{n}^2 \rangle - \langle \hat{n} \rangle^2 && (61)\\ &= \eta - \eta^2 && (62)\\ &= \boxed{\eta(1-\eta)}. && (63) \end{aligned}$$

To find the maximum, differentiate with respect to η:

$$\frac{d}{d\eta}\left[\eta(1-\eta)\right] = 1 - 2\eta. \tag{64}$$

Setting this equal to zero gives

$$\eta = \frac{1}{2}. \tag{65}$$

Since

$$\frac{d^2}{d\eta^2}\left[\eta(1-\eta)\right] = -2 < 0, \tag{66}$$

this is a maximum. The maximum variance is

$$\mathrm{Var}(\hat{n})_{\max} = \frac{1}{4}. \tag{67}$$

At $\eta = 0$, the output is always the vacuum state, so the photon number is certainly zero. At $\eta = 1$, the photon is always transmitted, so the photon number is certainly one. The variance vanishes in both cases because there is no uncertainty in the measurement outcome.